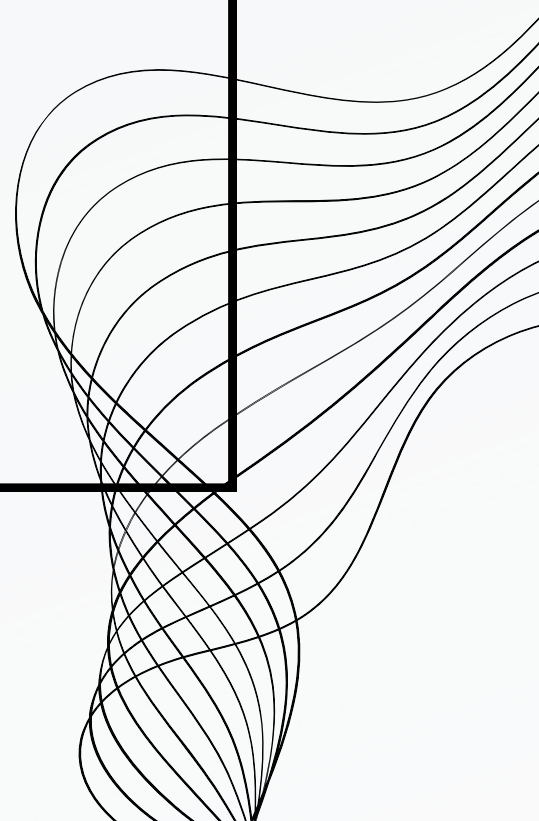




FIT3164 MDS8

FINAL PRESENTATION

**VISUAL PERCEPTION FROM A
DRONE'S PERSPECTIVE**

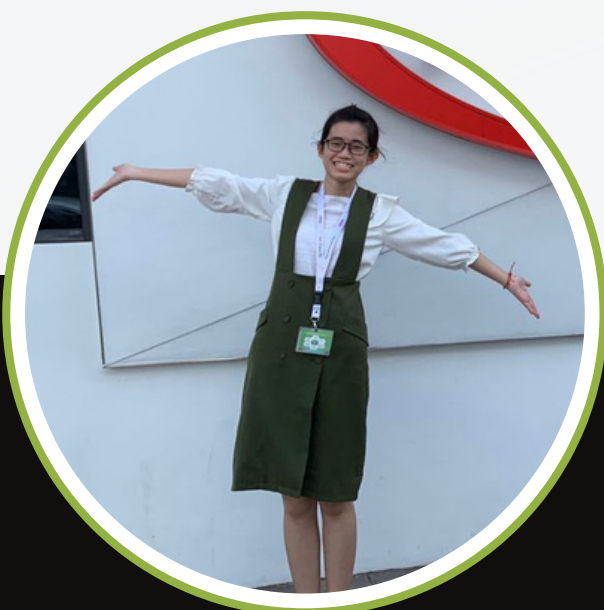


MEET OUR TEAM



Fong Chun Kent

Project Manager



Choo Jing Rou

Technical Lead



Emma Chong
Yong Yong

QA Lead



Prof. Raphael
Phan

Project
Supervisor



CONTENT



01

INTRODUCTION

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POSSIBLE IMPROVEMENTS

INTRODUCTION



- Drones are a type of flying robot that can be flown autonomously using software-controlled flight plans.
- Usage: surveillance, delivery



- **Human action recognition**
- Perform video analysis on human actions, based on drone-captured videos



OBJECTIVES & DELIVERABLES

Objectives

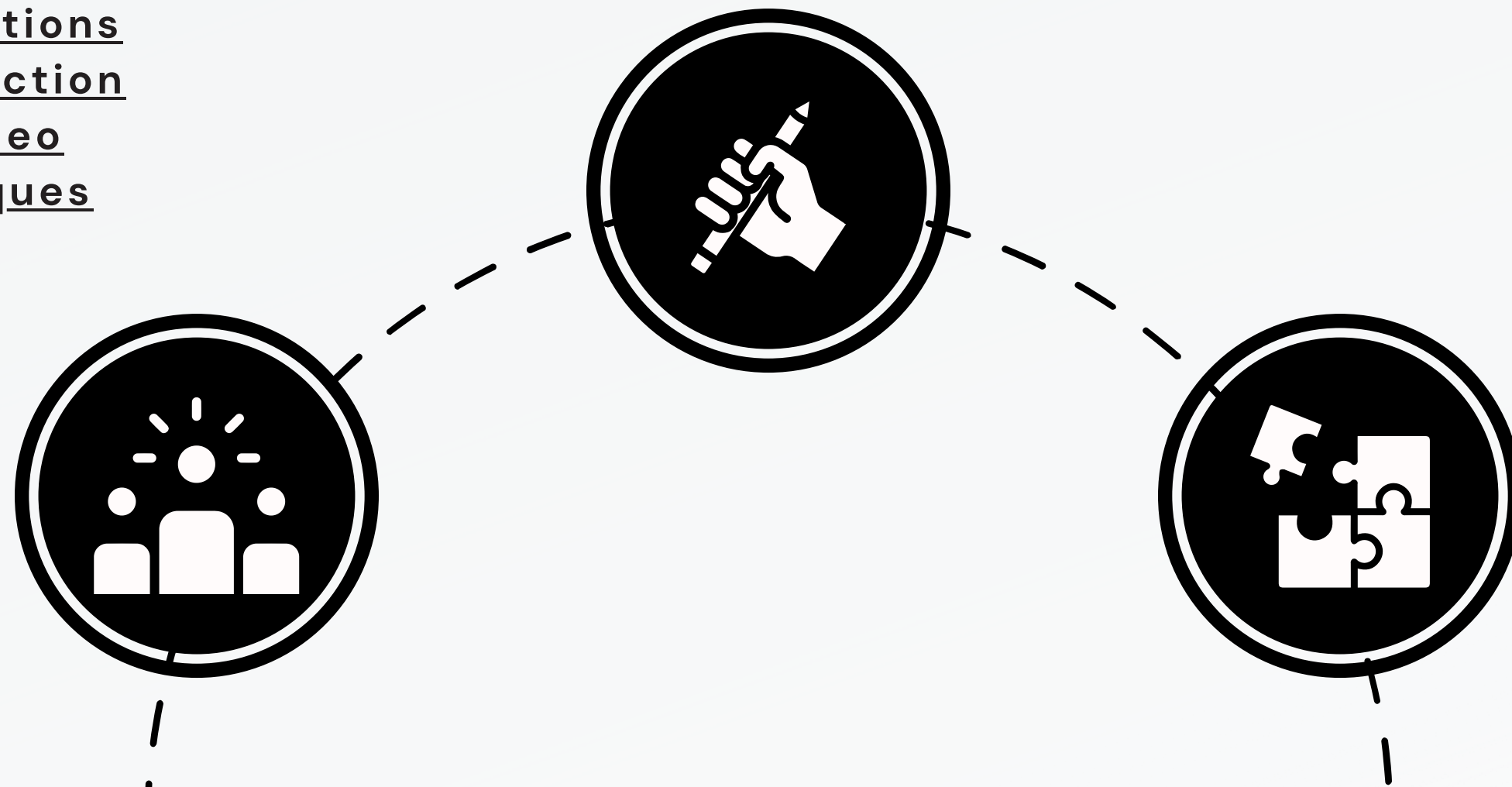
Design a new advanced model to improve the baseline results of the existing open source code implementations (DNN) of human action recognition video analysis techniques

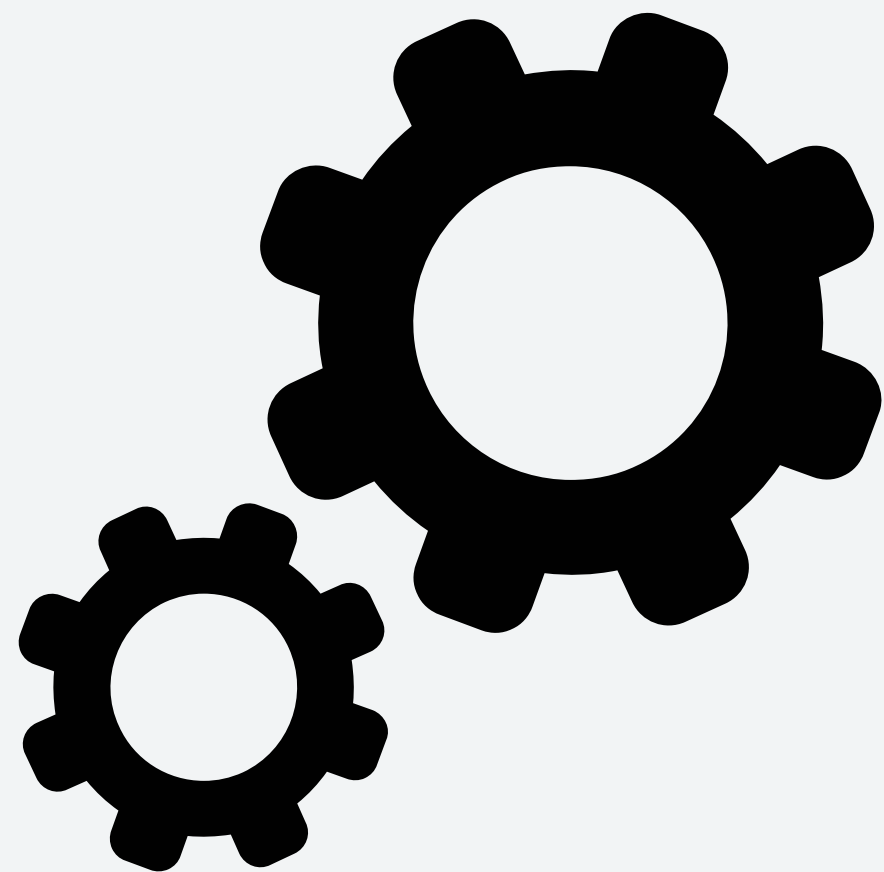
Deliverable 1

Develop advanced versions of the baseline implementations as prototypes to demonstrate the project objectives

Deliverable 2

Evaluate the performance of the prototype on various real-world settings and/or datasets





PROJECT MANAGEMENT

- Software Development Methodology
- Project Management Tools
- Task Allocation

PROJECT MANAGEMENT TOOLS

COMMUNICATION



DOCUMENTATION



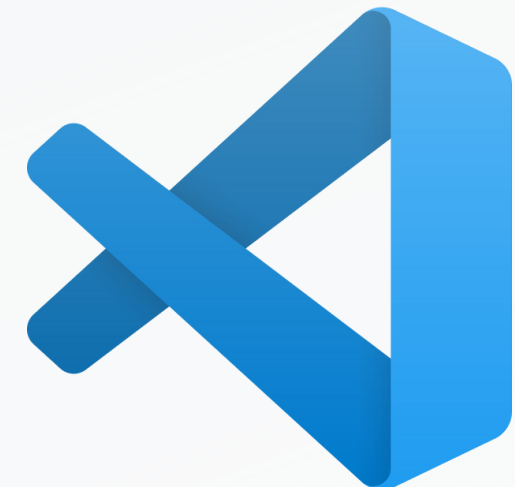
FILE STORAGE



REMOTE DESKTOP



CODE EDITOR



TASK ALLOCATION

Fong Chun Kent



Project Manager

- Schedules team meetings
- Ensures that the project is on the right track and the progress is on schedule
- Assists in technical development

Choo Jing Rou



Technical Lead

- Focuses on the project's technical developments
- Researches on different deep learning models/library
- Performs tasks like model training/testing, error handling

Emma Chong Yong Yong



QA Lead

- Ensure that the project deliverables are achieved
- Proof reads documents to ensure that they satisfy the specification requirements before submission
- Writes meeting minutes

SOFTWARE DEVELOPMENT METHODOLOGY

WHAT?

Agile approach is more suitable for our project

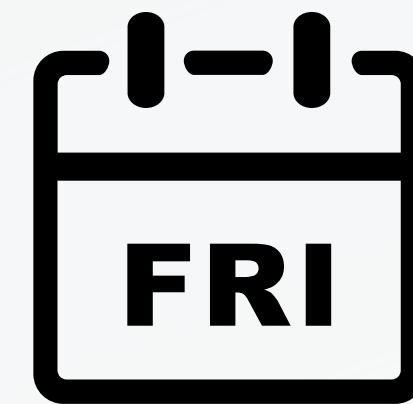
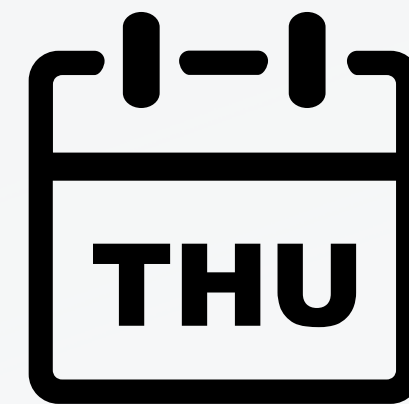
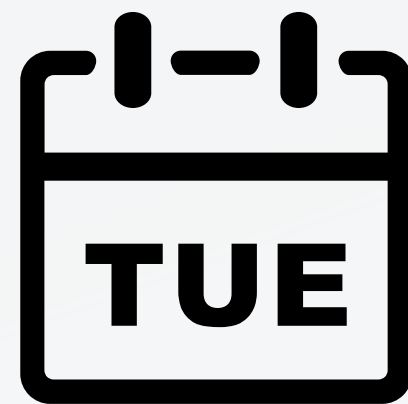
WHY?

Major changes of direction and scopes are widely common, we can sync with any real time changes that may arise during the product's life cycle

HOW?

- Capture all details with extensive documentation
- Frequent team meetings
- Sprint based work schedule

WEEKLY SPRINT WORKFLOW



MONDAY

Individual
work/progress

TUESDAY

Individual
work/progress

WEDNESDAY

Project manager
conducts follow-ups
with each team
member

THURSDAY

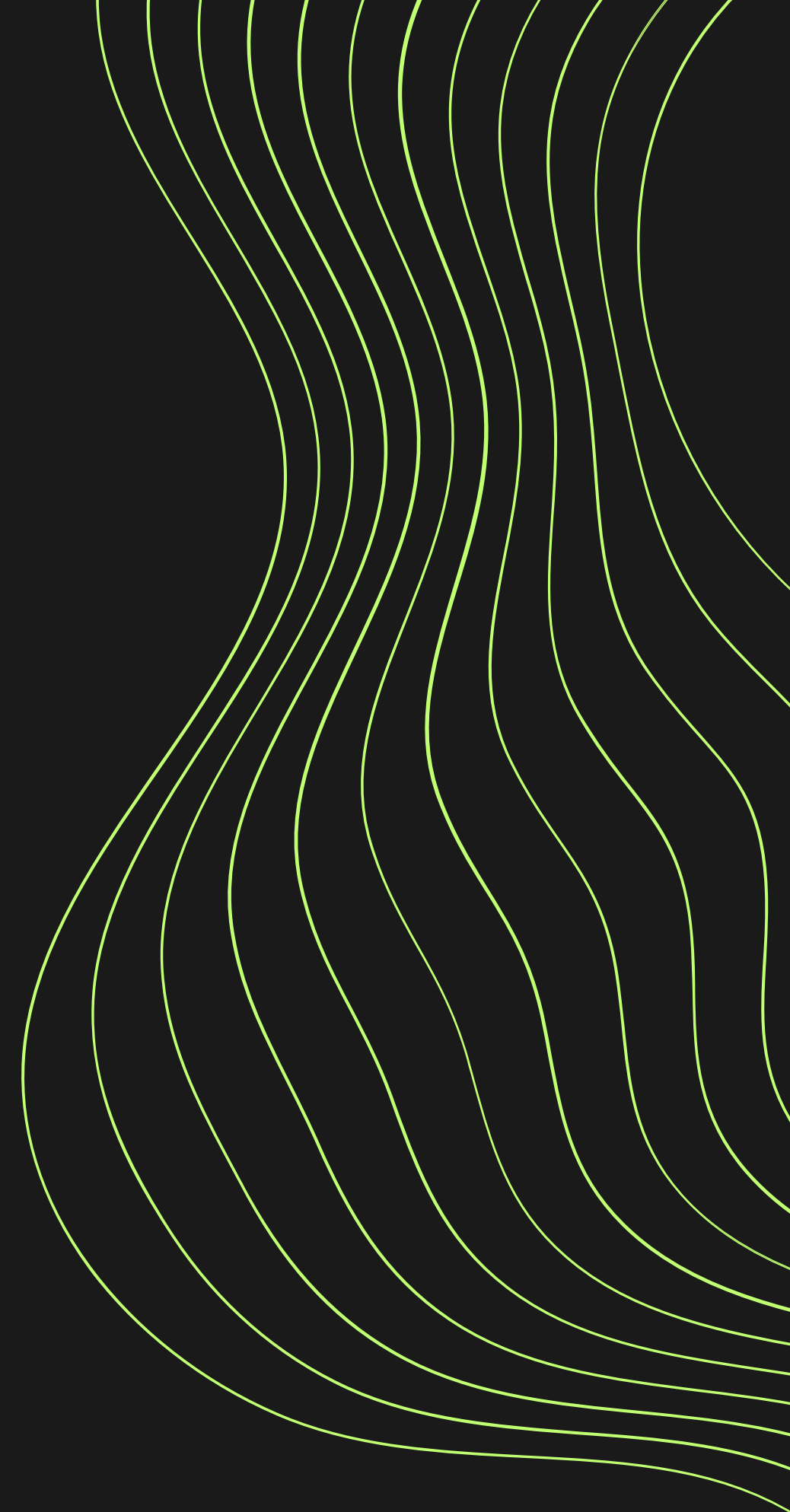
Meeting to plan next
week's tasks

FRIDAY

- Finalize and update our weekly tasks
- Submit assignments



PROJECT METHODOLOGY

- Specification
 - Training Phase
 - Models Used
- 

SPECIFICATION



- Widely known usability
- Chosen for its ease of use and powerful packages



- basic working blocks to compute and work with multi-dimensional arrays and matrices.
- utilise numpy for numerical representation of deep learning networks,



- Better processing speed
- More extensible framework

MODELS USED

<u>DNN</u> • Artificial neural network with 4 layers between the input and output layers	<u>Multi Layer Perceptron (MLP).</u> • a fully connected class of feedforward artificial neural network, with hidden layer sizes (20, 30, 40)
<u>Quadratic Discriminant Analysis</u> • A generative model, assumes each class follow a Gaussian distribution.	<u>XGBoost</u> • Implements machine learning algorithms under the Gradient Boosting framework with max depth 8 and 500 estimators
<u>VGG Origin/mobilenet_thin</u> • OpenPose pretrained models for pose estimation • Uses convolutional layers to extract features from images.	<u>COCO model</u> • neural network-based approach for detecting and localizing keypoints of human body joints in videos • Used in data preprocessing step

TRAINING PROCESS

Joints Tracking

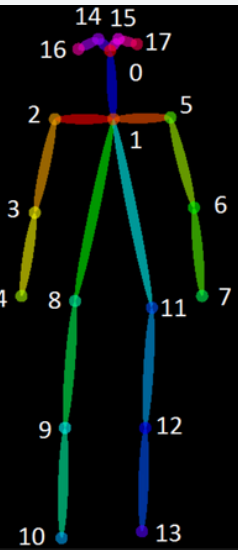
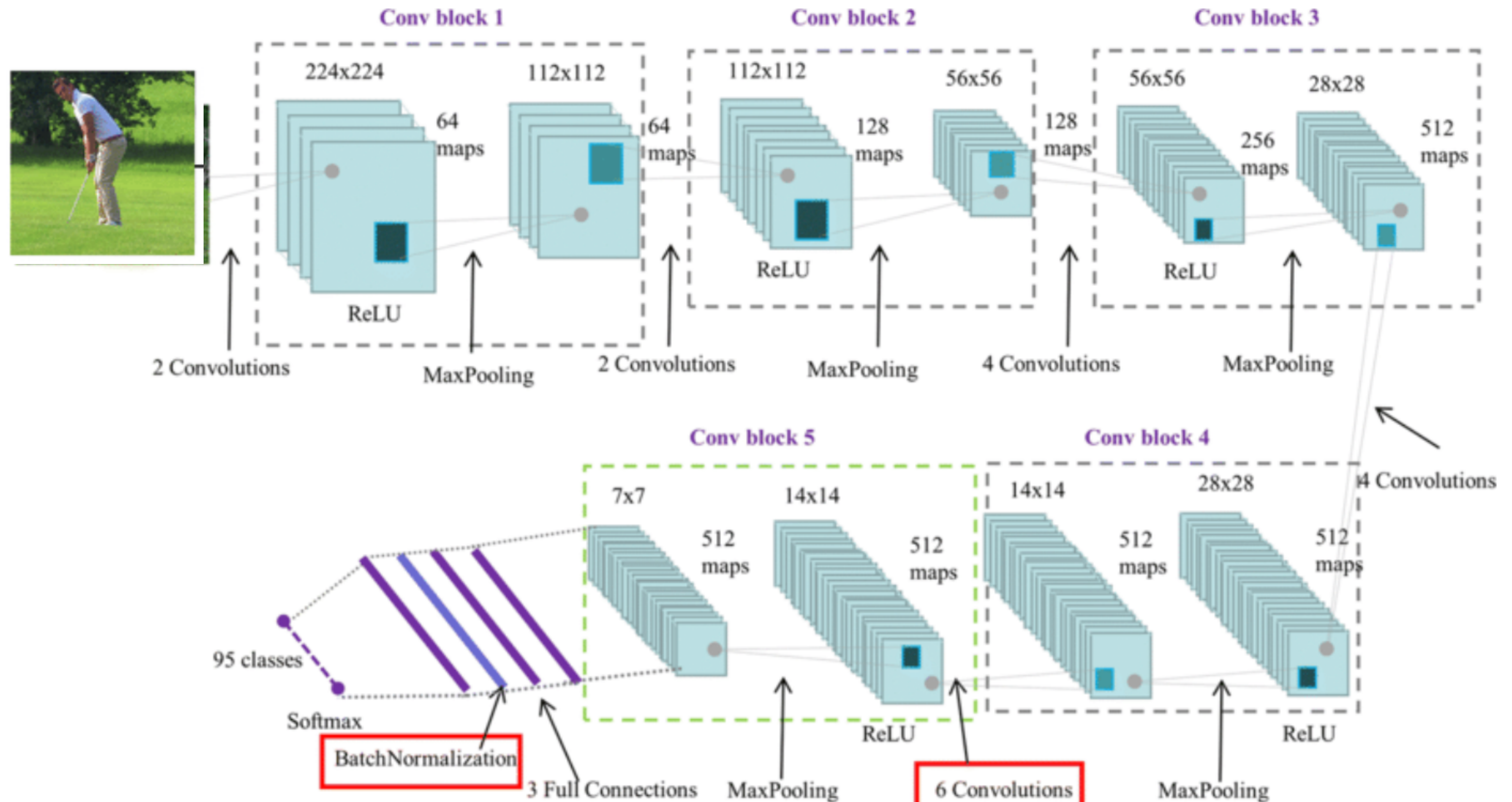
- OpenPose COCO mode
 - Deepsort Algorithm
- Detecting and localising keypoints of human body joints in videos

Pose Estimation

- VGG_origin
 - mobilenet_thin
- Feature extraction and classification into pose

Action Recognition

- Deep Neural Network
- Multi Layer Perceptron
- XGBoost
- Quantitative Discriminant Analysis



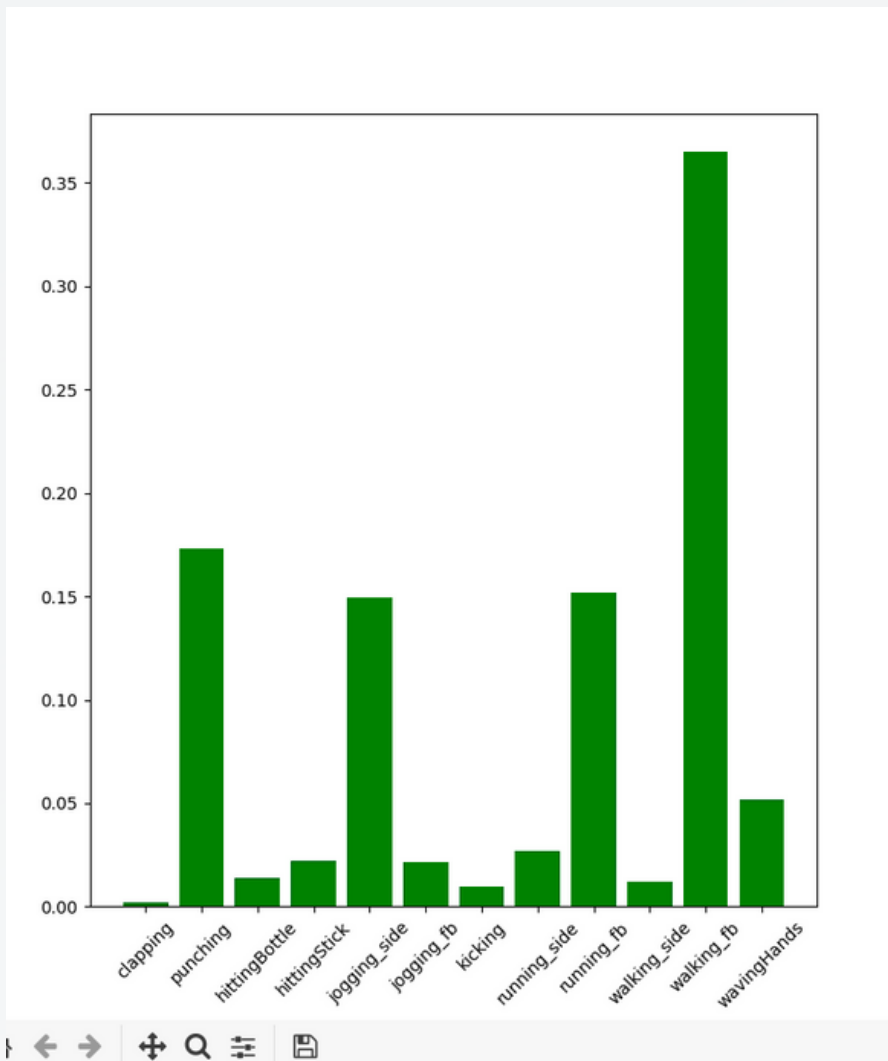
SOFTWARE DELIVERABLES

- Expected Output
- Results
- Satisfied Requirements
(Project Outcomes)



DELIVERABLES

Demonstration Video + Real Time Analysis



- Utilises Matplotlib and OpenCV to visualize the probability of prediction of each class
- Usage of Pose Estimator and Action Recognition models for prediction

Joint Coordinates Converter

```
import os
from os.path import exists, join, basename, splitext

videos = []
for video in os.listdir(input_directory):
    InputVideo = input_data + video
    folder = output_directory
    !mkdir {folder}
    videos.append(video)
    print(video)

!cd openpose && ./build/examples/openpose/openpose.bin --video {InputVideo}
--write_json {folder} --display 0 --render_pose 0 --model_pose COCO

print("Number processed of videos: ", videos.__len__())
```

- Utilises the COCO model to predict joints and convert to x,y coordinates for training purpose

An abstract graphic featuring a dark background with white, flowing, wavy lines that create a sense of movement and depth. The lines are thin and delicate, curving and looping across the frame. The central text 'INPUT AND OUTPUT' is rendered in a bold, white, sans-serif font, positioned horizontally in the middle of the image. The overall aesthetic is modern and minimalist.

INPUT AND OUTPUT

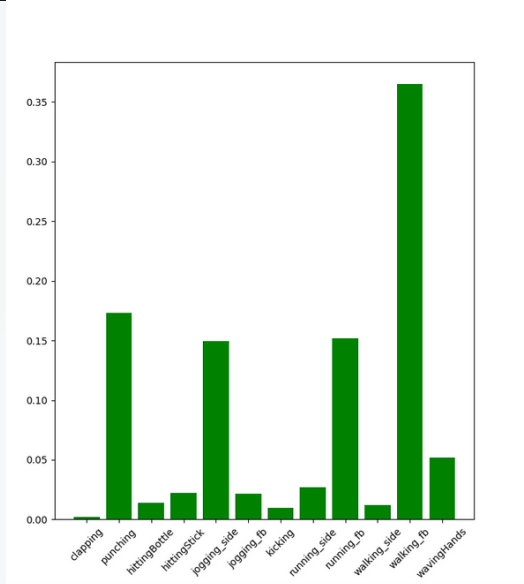
```
main.py
```

Video of Human Action

A person wearing a blue long-sleeved shirt, black pants, and a yellow beanie is running on a dirt field. They are holding a white disc in their right hand. The background consists of a dirt field and a line of tall, dry grass.

OpenPose-
COCO.ipynb

Classified Pose + Action

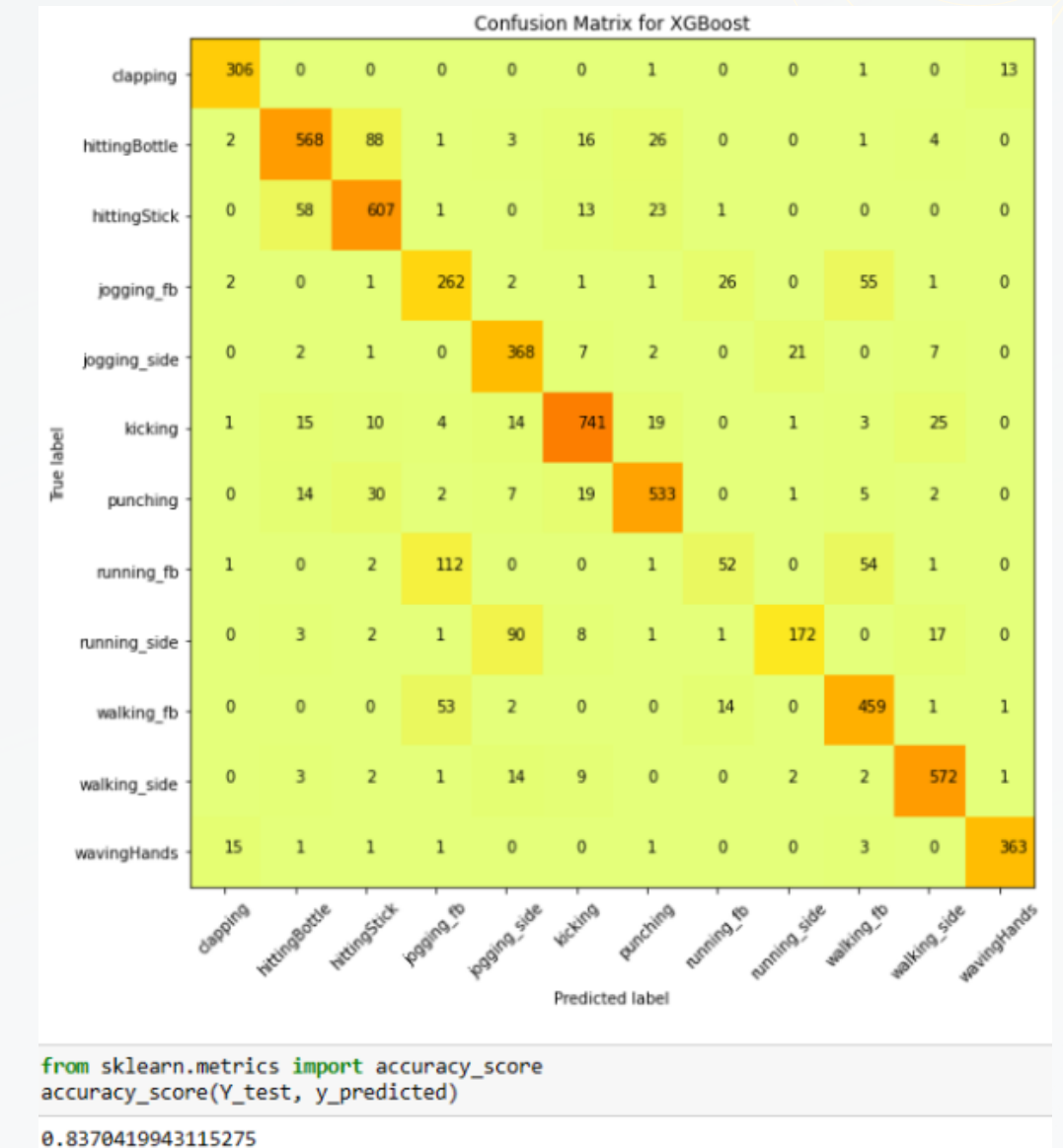


CSV file with x,y coordinates of joints

AX	AY	AZ	BA	BB	BC
REar_y	REar_c	LEar_x	LEar_y	LEar_c	class
0	0	0.472492708333333	0.444556481481481	0.902129	punching
0	0	0.4739765625	0.444756481481481	0.874362	punching
0	0	0.475480729166667	0.444732407407407	0.907749	punching
0	0	0.478541666666667	0.447357407407407	0.887322	punching
0	0	0.475494270833333	0.447374074074074	0.918557	punching
0	0	0.480042708333333	0.444782407407407	0.879086	punching
0	0	0.480055208333333	0.447387962962963	0.914204	punching
0	0	0.481620833333333	0.450087962962963	0.961605	punching
0	0	0.483046875	0.450162962962963	0.937832	punching
0	0	0.4800953125	0.450180555555556	0.91763	punching
0	0	0.4800765625	0.452837037037037	0.906149	punching
0	0	0.478529166666667	0.450156481481481	0.875922	punching
0	0	0.4785484375	0.450116666666667	0.918992	punching
0	0	0.478565104166667	0.45007962962963	0.916163	punching
0	0	0.481565104166667	0.447515740740741	0.952551	punching
0	0	0.4800640625	0.447340740740741	0.902173	punching
0	0	0.4800828125	0.444771296296296	0.930833	punching
0	0	0.475453125	0.447331481481481	0.931695	punching

SATISFIED REQUIREMENTS

Title	Functional / Non-Functional	Description
Improved classifier accuracy	Functional	Improved from 73.62% to 83.70%.
Discover applicable datasets	Functional	Two datasets were found and processed.
Save analysed videos	Non-Functional	Videos are saved in new .mp4 file.



S02_wavingHands_HD.mp4

40.2 MB



S01_wavingHands_HD_tf_out...

30.0 MB

POSSIBLE IMPROVEMENTS



LIMITATIONS



LIMITED ACTIONS

- The current models can only recognize 12 actions



ONE ACTION PER FRAME

- Unable to identify > 1 human actions in the same video frame

POSSIBLE IMPROVEMENTS

- Utilise RNN(LSTM) model to classify actions with dynamic joints.
- Usage of Spatial Transformer Networks to enhance spatial invariance and geometric features of input data
- Work on wider range of datasets to overcome overfitting

THANK YOU